

Mechanism	Primary purpose
SELinux	Label-based access control
AppArmor	Path-based access control
Seccomp	System call filtering
Namespaces	Resource isolation
Capabilities	Least-privilege execution
Landlock	Filesystem sandboxing
Containers/VMs	Workload isolation
Nvidia OpenShell	Runtime governance for AI agents

Traditional Linux mechanisms focus primarily on processes and resources. OpenShell adds AI-specific runtime controls on top of those foundations rather than replacing them.

OpenShell should not be viewed as a replacement for SELinux, AppArmor, seccomp, namespaces, containers, or virtual machines. These mechanisms continue to provide the fundamental isolation boundaries of Linux systems. OpenShell operates one layer above them, providing runtime governance tailored to AI agents. In practice, it complements existing security primitives rather than replacing them.

Traditional Linux security mechanisms focus primarily on processes and resources. OpenShell addresses additional concerns such as controlled access to files, network communication, credential isolation, and interactions with model providers. It builds on top of existing operating system capabilities rather than introducing a new isolation model.

AI agents introduce a different execution model from traditional applications. Nvidia OpenShell builds upon established Linux security primitives rather than replacing them. By combining sandboxing, policy-based controls, privacy-aware inference routing, and kernel-enforced restrictions, it provides a framework for governing autonomous AI workloads while leveraging the proven foundations already available in Linux. Rather than reinventing isolation mechanisms, OpenShell demonstrates how existing Linux security capabilities can be combined with additional runtime controls to address the unique requirements of AI agents. **END** 🐧

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